O-S-639B
August 27, 1984
SUPERSEDING
O-S-639A
August 9, 1961
(See 6.6)

FEDERAL SPECIFICATION

SODIUM PHOSPHATE, DIBASIC, ANHYDROUS, TECHNICAL

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers dibasic sodium phosphate suitable for water softening purposes and as an ingredient in compounded metal cleaners.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-E-485 - Enamel, Semigloss, Rust-Inhibiting.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-723 - Drums, Fiber.

Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 6810

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(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Federal Regulation:

Department of Transportation (DOT)

49 CFR 171-178 - Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

D 460 - Sampling and Chemical Analysis of Soap and Soap Products.
D 820 - Chemical Analysis of Soaps Containing Synthetic Detergents.
D 1193 - Reagent Water.
D 3951 - Standard Practice for Commerical Packaging.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA.)

(Industry association specifications and standards are generally available for references from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. Anhydrous dibasic sodium phosphate (Na_2HPO_4) shall be in the form of a white powder, flakes or granules.

3.2 pH value. The pH of a one percent solution shall be between 8.7 and 9.3 (see 4.5.2).

3.3 Composition. The composition shall be in accordance with table I.

TABLE I. Composition.

	Maximum percent	Minimum percent	Test
Na_2HPO_4		97.5	4.5.3.1
Water	2.0	----	4.5.3.2
Chlorides as NaCl	0.2	----	4.5.3.3
Insoluble matter	0.15	----	4.5.3.4

3.4 Material safety data sheet. The contracting activity shall be furnished a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD Form OSHA-20 and found as part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lot. A lot shall consist of material from the same batch or blending operation. If a continuous process is used, a lot shall consist of all material subjected to the same processing operations with no change in operating conditions for 1 day.

4.3 Sampling for tests. At least three specimens shall be selected for sampling in accordance with inspection level S-2 of MIL-STD-105. Both lot size and sample number shall be expressed as number of drums or cans. Each specimen shall be kept separate and tested separately as specified in 4.5. Each 0.5-pound sample shall be placed in an airtight container, which shall be almost completely filled by the sample. Each container shall then be sealed, marked, and accurately weighed. The weight and date of sampling shall be recorded on each container, and containers shall be forwarded to a laboratory satisfactory to the contracting activity for testing.

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4.4 Examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level I. An acceptable quality level (AQL) of 1.5 percent defective shall be used to verify compliance with all stipulations of this specification regarding fill, closure, marking and other requirements not involving tests. Containers shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or under required fill shall be rejected. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan in accordance with MIL-STD-105, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 Tests. Any tests requiring time in excess of 4 hours shall be performed on only one-third of the specimens; all other tests shall be made on each specimen to provide information on variability.

4.5.2 pH value. The pH value of a one percent solution in boiled distilled water in accordance with type III of ASTM D 1193 shall be determined (the water having a pH value between 6.0 and 7.0) by electrometric means at 25 degrees Celsius (°C) using a glass/reference electrode system.

4.5.3 Composition.

4.5.3.1 Disodium phosphate. The test shall be conducted as specified in 4.5.3.1.1 except that in case of dispute the referee test as specified in 4.5.3.1.2 shall be used.

4.5.3.1.1 Composition test. A 2.5 ± 0.001 gram sample shall be transferred to a 400 milliliter (mL) beaker. Dissolve in 100 mL of distilled water. Add 15 mL concentrated hydrochloric acid (HCl) (specific gravity 1.19). Boil the covered solution for 30 minutes. Cool to room temperature, and dilute to 200 mL. Neutralize to 4.3 pH as measured by a pH electrometer and glass/reference electrode system, using 50 percent sodium hydroxide (NaOH) solution (approximately 50 grams per 100 mL solution) until reaching about 3.5 pH, and making the final adjustment with standard NaOH solution (0.5N). Then titrate to 8.8 pH with the standard 0.5N NaOH solution, recording the titration between 4.3 and 8.8 pH. The electrometer shall be standardized against a pH 7 buffer solution immediately before use.

Calculation:

$$\text{Percent } Na_2HPO_4 = \frac{M \times N \times 14.196}{W}$$

where:

W = weight of sample.
M = milliliters of NaOH solution used between 4.3 and 8.8 pH.
N = normality of NaOH solution.

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4.5.3.1.2 Referee test. Weigh, to the nearest 0.1 milligram (mg), approximately 0.1 gram of the sample. Dissolve in distilled water and make up to 250 mL in a volumetric flask. Take a 50 mL aliquot of the solution and determine the phosphate in accordance with method 21-28 of ASTM D 820.

4.5.3.2 Water. The water content shall be determined in accordance with method 11 of ASTM D 460.

4.5.3.3 Chlorides. A 0.2 gram sample shall be dissolved in 20 mL of distilled water and 3 mL of 7 N nitric acid and 1 mL of 0.1 N silver-nitrate solution added. Any turbidity shall be not greater than is produced by a solution of 0.002 percent sodium chloride of an equal volume with the same quantities of acid and silver-nitrate solution.

4.5.3.4 Insoluble matter. Dissolve 2.5 grams of sample in 500 mL of warm (60 to 65°C) distilled water. If turbidity or undissolved matter is present, filter through a weighed Gooch crucible, washing with five 10 mL portions of warm distilled water. Dry the crucible and contents to a constant weight by heating at 100 to 105°C. Weigh and calculate the percentage of insoluble matter.

$$\text{Percent of insoluble} = \frac{\text{grams of residue}}{\text{grams of sample}} \times 100$$

4.6 Inspection of packaging. (See 4.4)

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Sodium phosphate shall be furnished in 1-pound cans or 25-pound, 150-pound or 325-pound drums, as specified (see 6.2).

5.1.1.1 Cans. One-pound quantities of sodium phosphate shall be preserved-packaged in cans in accordance with type V, class 2 of PPP-C-96, exterior coating plan B, with side seam stripping.

5.1.1.2 Drums. The 25-pound, 150-pound and 325-pound quantities shall be furnished in drums specified in 5.2, as applicable.

5.1.2 Commercial. Sodium phosphate shall be preserved in accordance with ASTM D 3951 and shall meet the requirements of DOT regulations, title 49.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cans. One-pound cans of sodium phosphate shall be arranged and packed for overseas shipment in accordance with the appendix to PPP-C-96 except that only one tier of 24 cans shall be packed in a box and V3c fiberboard is an acceptable alternate to V2s for exterior containers. Boxes shall be closed, reinforced and waterproofed in accordance with the applicable box specification or appendix thereto.

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5.2.1.2 Fiber drums. The 25-pound and 150-pound quantities of sodium phosphate shall be packed in fiber drums in accordance with type III, grade A of PPP-D-723. Drums shall be provided with a polyethylene heat-sealed bag type liner having a minimum thickness of 3 mils. Excess air shall be brought to a practicable minimum within the bag prior to sealing.

5.2.1.3 Metal drums. The 325-pound quantities of sodium phosphate shall be packed in metal drums in accordance with 37A of DOT regulations, title 49. Drum exteriors shall be coated with olive drab enamel in accordance with TT-E-485. Drums shall be provided with liners as specified in 5.2.1.2.

5.2.2 Level B.

5.2.2.1 Cans. One-pound cans of sodium phosphate shall be arranged and packed for domestic shipment in accordance with the appendix to PPP-C-96 except that only one tier of 24 cans shall be packed in a box. Boxes shall be closed in accordance with the applicable box specification or appendix thereto.

5.2.2.2 Drums. Unless otherwise specified (see 6.2), 25-pound, 150-pound and 325-pound quantities of sodium phosphate shall be packed in accordance with 5.2.1.2 and 5.2.1.3, as applicable, except that type II, grade A of PPP-D-723 drums are acceptable.

5.2.3 Commercial. Sodium phosphate, preserved as specified (see 6.2), shall be packed in accordance with ASTM D 3951 and shall meet the requirements of DOT regulations, title 49.

5.3 Pallets. When specified (see 6.2), material shall be palletized in accordance with MIL-STD-147.

5.4 Marking.

5.4.1 Military agencies. In addition to any special marking required by the contract or order and the marking specified in 5.4.3, interior containers, exterior shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129, ASTM D 3951 and DOT regulations, title 49. Markings shall include the specification number.

5.4.2 Civil agencies. In addition to the marking specified in 5.4.3, shipments shall be marked in accordance with FED-STD-123, ASTM D 3951 and DOT regulations, title 49 as applicable.

5.4.3 Precautionary marking. Interior packs and shipping containers shall include the following precautionary marking:

"STORE IN A COOL PLACE.
KEEP TIGHTLY CLOSED."

5.5 Material safety data sheet. A copy of the material safety data sheet (Form OSHA-20) shall be attached to the shipping document for each destination (see 3.4).

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6. NOTES

6.1 Intended use. The dibasic sodium phosphate covered in this specification is primarily intended for use in water softening purposes and as an ingredient in compounded metal cleaners. The material is not intended for medicinal purposes. Water softened with this material is not intended for human consumption.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in acquisition documents:

(a) Title, number and date of this specification.
(b) Pound quantity per unit container required (see 5.1).
(c) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).
(d) Level B requirements for drums, if different (see 5.2.2.2).
(e) Whether palletization is required (see 5.3).

6.3 Dibasic sodium phosphate is also known commerically as disodium phosphate, sodium hydrogen phosphate, and disodium orthophosphate.

6.4 Transportation description. Transportation description applicable to these items is:

Disodium-phosphate.
Less carload or less truckload, in cloth or 5-ply paper bags, or in barrels or boxes; carload or truckload, in packages.
Carload weight 40,000 pounds.
Truckload weight 40,000 pounds.

6.5 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ABCA-NAVSTAG 6E. When amendment, revision or cancellation of this specification is proposed which affects or violates the international agreement concerned. the preparing activity will inform GSA so that appropriate reconciliation action may be taken through international standardization channels.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - SH
Air Force - 68

Review Activities

DLA - GS

User Activities

Navy - AS

CIVIL AGENCY COORDINATING ACTIVITIES:

FSS
NBS
OSS

PREPARING ACTIVITY:
Navy - SH
(Project 6810-B408)

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